

Heavy Metals and Couple Fecundity, the LIFE Study

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The effect of heavy metals at environmentally relevant concentrations on couple fecundity has received limited study despite ubiquitous exposure. In 2005–2009, couples (n=501) desiring pregnancy and discontinuing contraception were recruited and asked to complete interviews and to provide blood specimens for the quantification of cadmium ($\mu\text{g/L}$), lead ($\mu\text{g/dL}$) and mercury ($\mu\text{g/L}$) using inductively coupled plasma-mass spectrometry. Couples completed daily journals on lifestyle and intercourse along with menstruation and pregnancy testing for women. Couples were followed for 12 months or until pregnant. Fecundability odds ratios (FORs) and 95% confidence intervals (CIs) were estimated adjusting for age, body mass index, cotinine, and serum lipids in relation to female then male exposures. FORs <1 denote a longer time to pregnancy. In adjusted models, reduced FORs were observed for both female cadmium (0.78; 95% CI 0.63–0.97) and male lead (0.85; 95% CI 0.73–0.98) concentrations. When jointly modeling couples' exposures, only male lead concentration significantly reduced the FOR (0.82; 95% CI 0.68, 0.97), though the FOR remained <1 for female cadmium (0.80; 95% CI 0.64, 1.00). This prospective couple based cohort with longitudinal capture of time to pregnancy is suggestive of cadmium and lead's reproductive toxicity at environmentally relevant concentrations.